



Global and Regional Market Research Report

December 2022



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Global CTA Industry Scenario





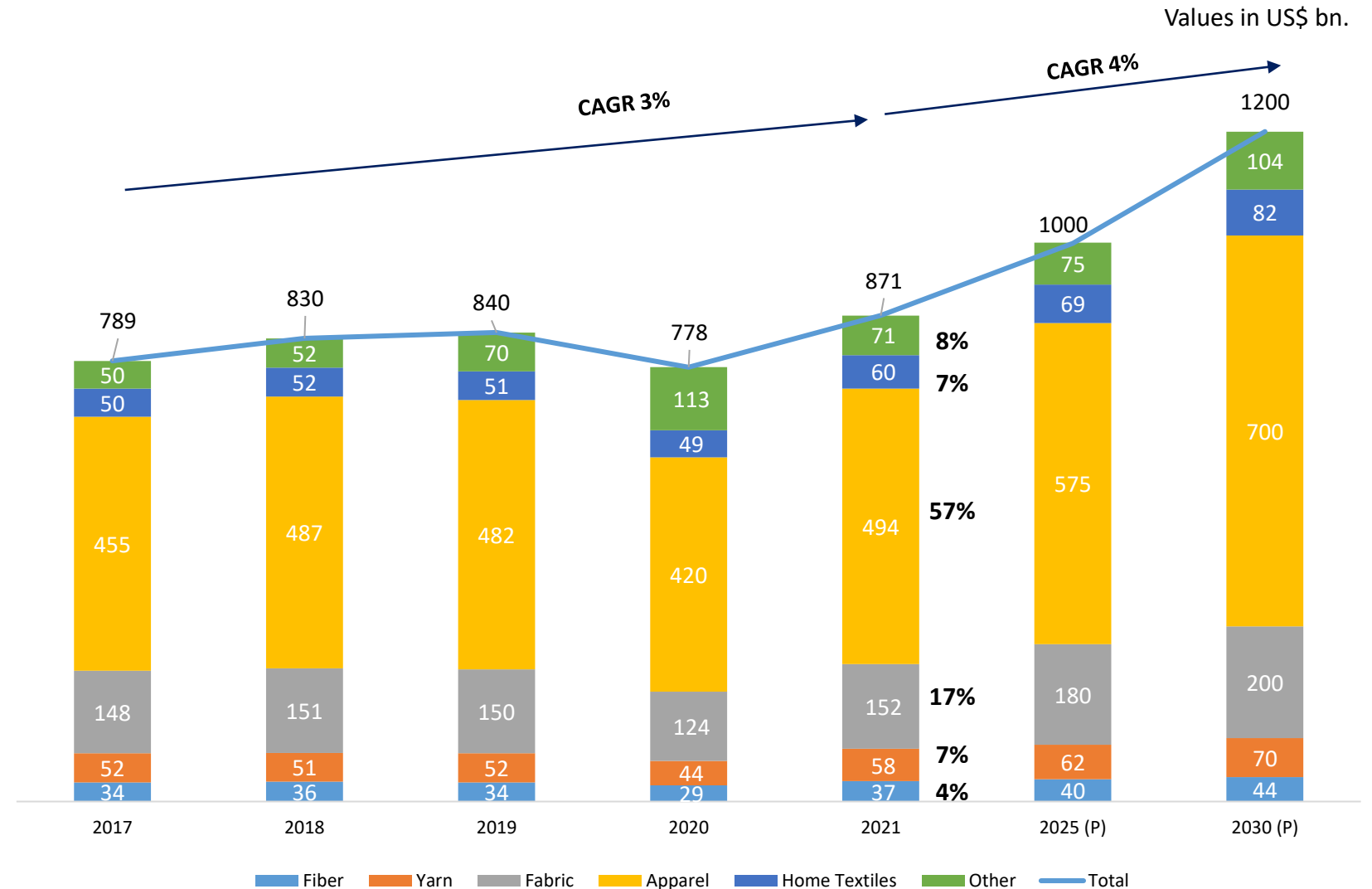
Export and Import Scenario



The global CTA exports will reach US\$ 1.2 Trillion by 2030

The global CTA exports have grown at a CAGR of 3% from 789 billion in 2017 to 871 billion in 2021. These are further expected to grow at a CAGR of 4% to reach US\$ 1.2 trillion in 2030.

Apparel is the largest traded commodity over the years, accounting for 57% of the total exports in 2021.



Source: UN Comtrade

Leading Exporters

Data for 2021

	China	US\$ 318 Bn	37%		Italy	US\$ 37 Bn	4%
	Bangladesh	US\$ 43 Bn	5%		Turkey	US\$ 34 Bn	4%
	India	US\$ 41 Bn	5%		USA	US\$ 26 Bn	3%
	Germany	US\$ 40 Bn	5%		Spain	US\$ 21 Bn	2%
	Vietnam	US\$ 38 Bn	4%		Netherlands	US\$ 20 Bn	2%

Source: UN Comtrade

US and EU collectively constitute approx. 45% of the global apparel market

Global apparel market size was approx. US\$ 1.5 trillion in 2021. The largest consumer country was USA with a market worth US\$ 257 billion, followed by EU and China.

The global apparel market is expected to grow by a CAGR of 4% and reach about US\$ 2 trillion by 2025. China is expected to emerge as the biggest apparel market by 2025, growing at a CAGR of 11% to reach US\$ 340 billion.

Values in US\$ mn.

Region	2019	2020	2021	CAGR 2019-21	CAGR (P) 2019-25	2025 (P)
United States	235	176	257	5%	2%	265
EU-27	264	219	211	-11%	1%	280
China	184	166	188	1%	11%	340
Japan	101	81	78	-12%	0.5%	105
India	78	55	72	-4%	10%	135
Brazil	48	34	39	-10%	4%	60
Canada	25	20	22	-6%	7%	37
RoW	690	517	600	-7%	2%	780
World	1625	1269	1467	-5%	4%	2,002

Source: Published Data and Wazir Analysis

Leading Importers

Data for 2021

	USA	US\$ 128 Bn	15%		United Kingdom	US\$ 29 Bn	3%
	Germany	US\$ 57 Bn	7%		Italy	US\$ 27 Bn	3%
	China	US\$ 35 Bn	4%		Spain	US\$ 24 Bn	3%
	Japan	US\$ 33 Bn	4%		Netherlands	US\$ 23 Bn	3%
	France	US\$ 33 Bn	4%		Vietnam	US\$ 23 Bn	3%

Source: UN Comtrade

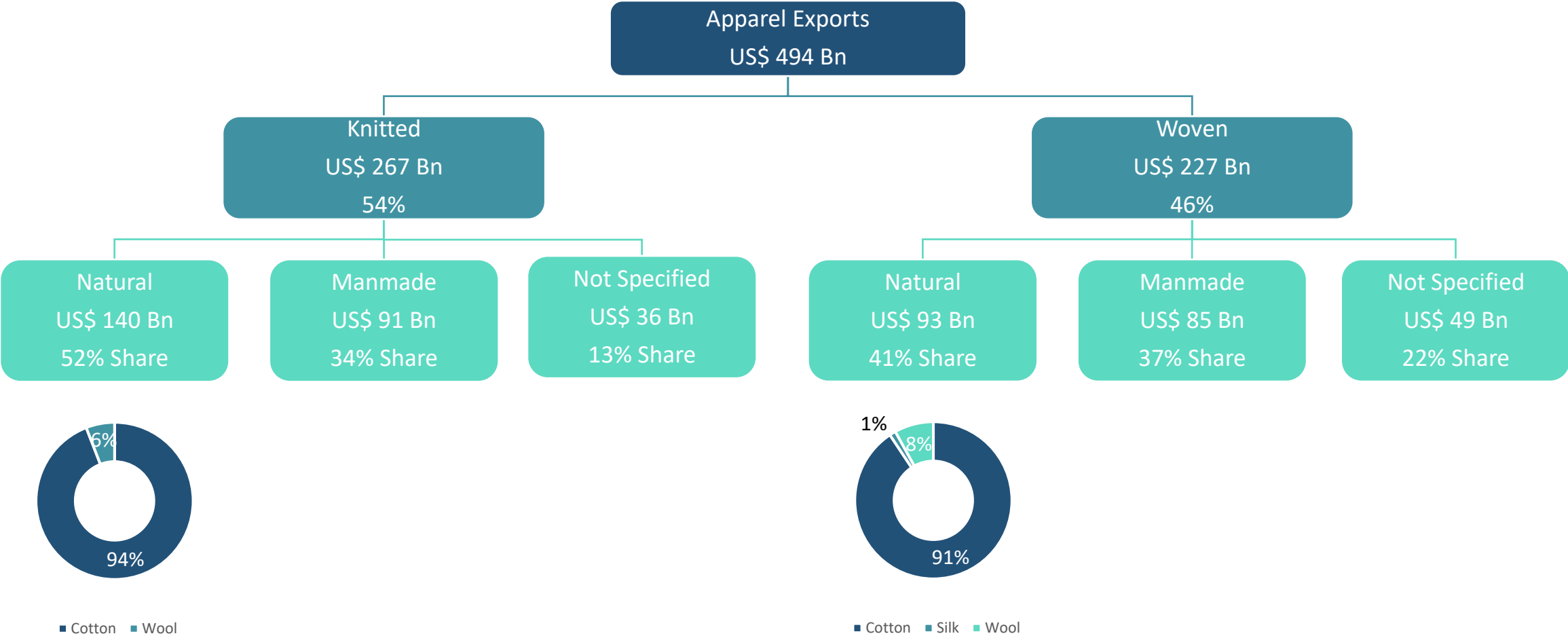


Cotton Landscape



Cotton knitted apparel are the highest exported commodity globally

Data for 2021



Source: UN Comtrade

The largest traded commodities globally are majorly cotton based

Data for 2021

HS Code	Description	Trade Value (US\$ Bn)	Trade Share
610910	Knitted t-shirts of cotton	33.7	4%
611020	Knitted jerseys of cotton	30.2	3%
611030	Knitted jerseys of manmade fibers	26.9	3%
620342	Woven men's trousers bib and brace overalls breeches and shorts of cotton	24.7	3%
630790	Made up articles of fibres other than natural and MMF	23.3	3%
620462	Woven women's trousers bib and brace overalls breeches and shorts of cotton	22.0	3%
520100	Raw cotton fibre	16.9	2%
610990	Knitted t-shirts of fibers other than cotton	14.2	2%
621210	Woven women's brassieres	11.8	1%
610462	Knitted women's trousers bib and brace overalls breeches and shorts of cotton	11.1	1%

Source: UN Comtrade

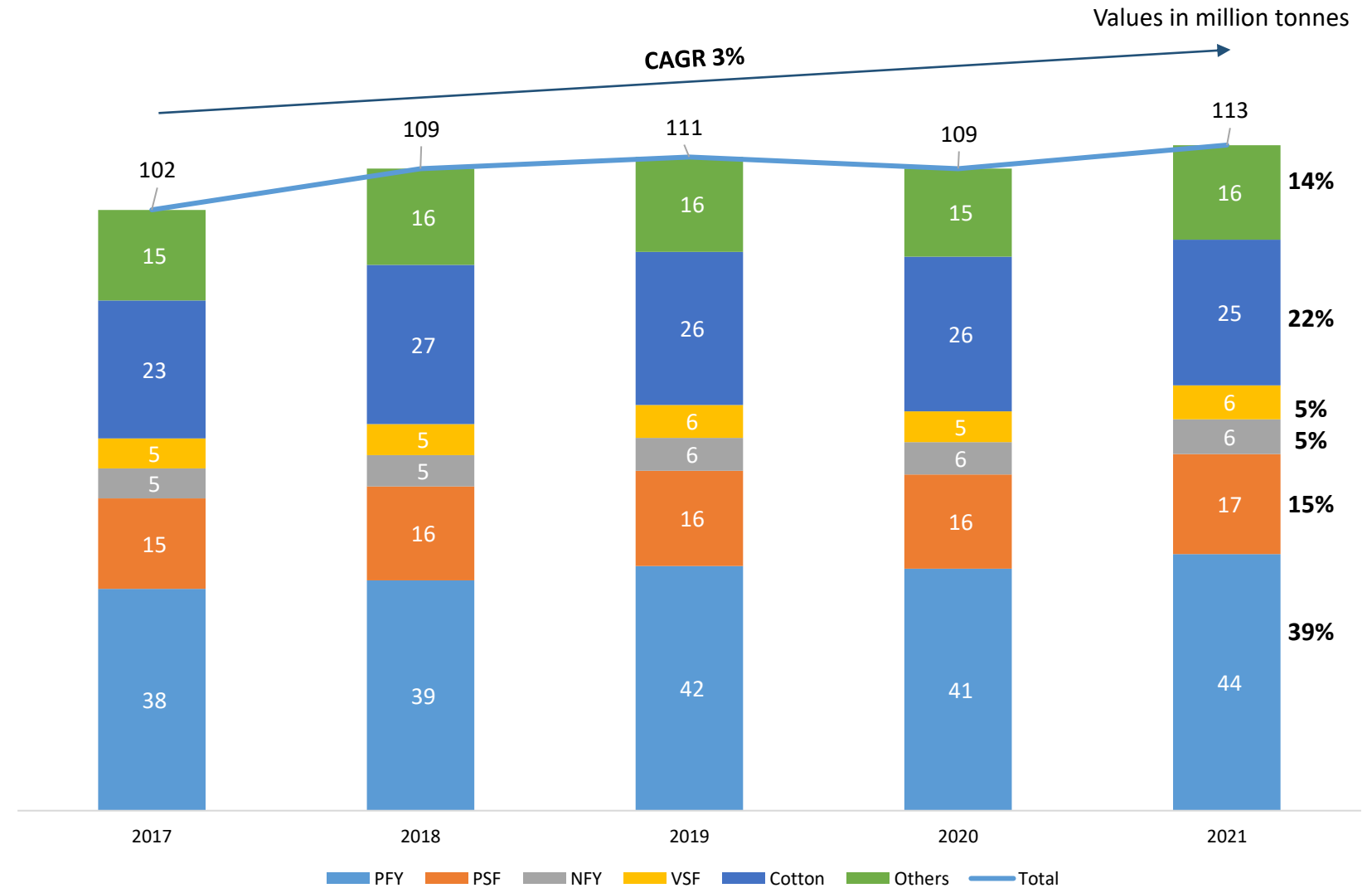
PFY fibre production has grown at a CAGR of 4% from 2017 to 2021

The global fiber production has seen a growth of 3% from 2017 to 2021.

PFY production has increased from 38 million tonnes in 2017 to 44 million tonnes in 2021, accounting for 39% of the global fiber production in 2021.

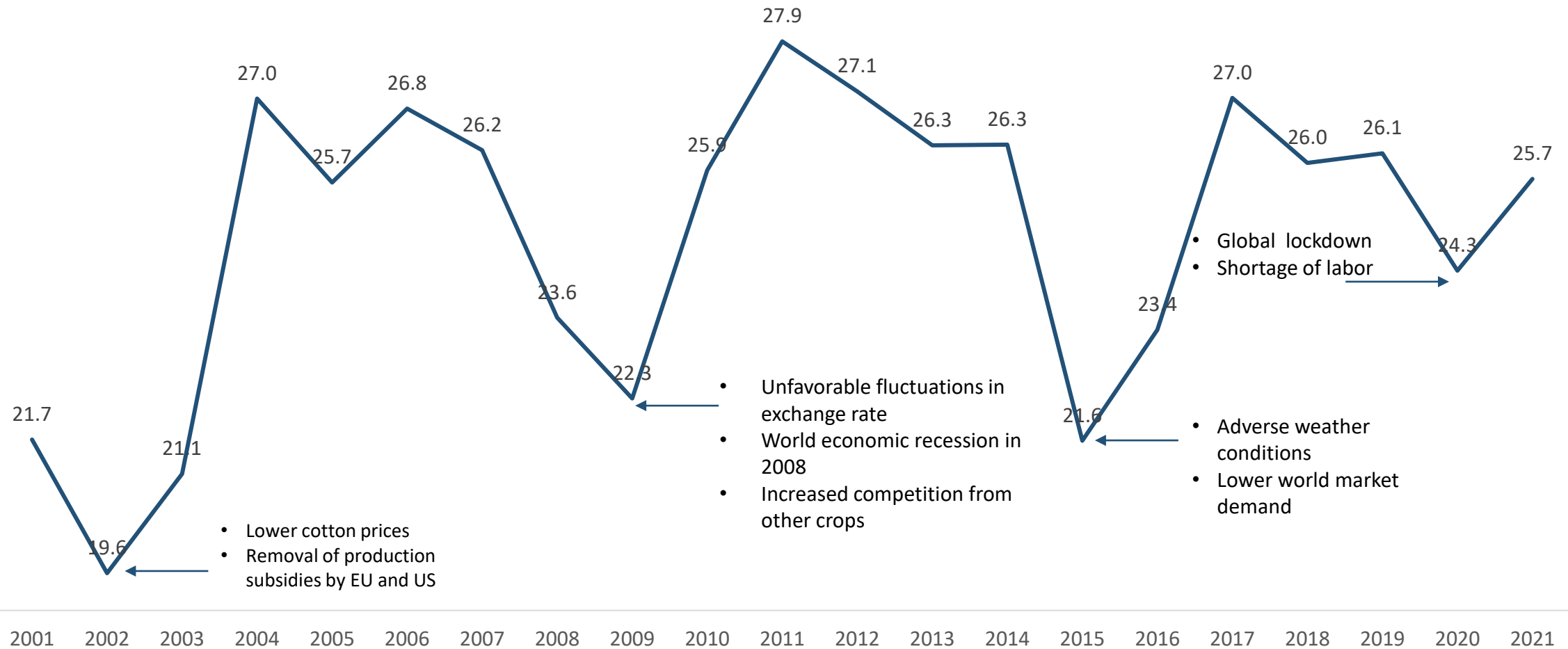
Cotton production has been approx. consistent throughout the years, growing at a CAGR of 2% from 2017 to 2021.

Source: Textile Exchange



Over the years, cotton production has seen major fluctuations

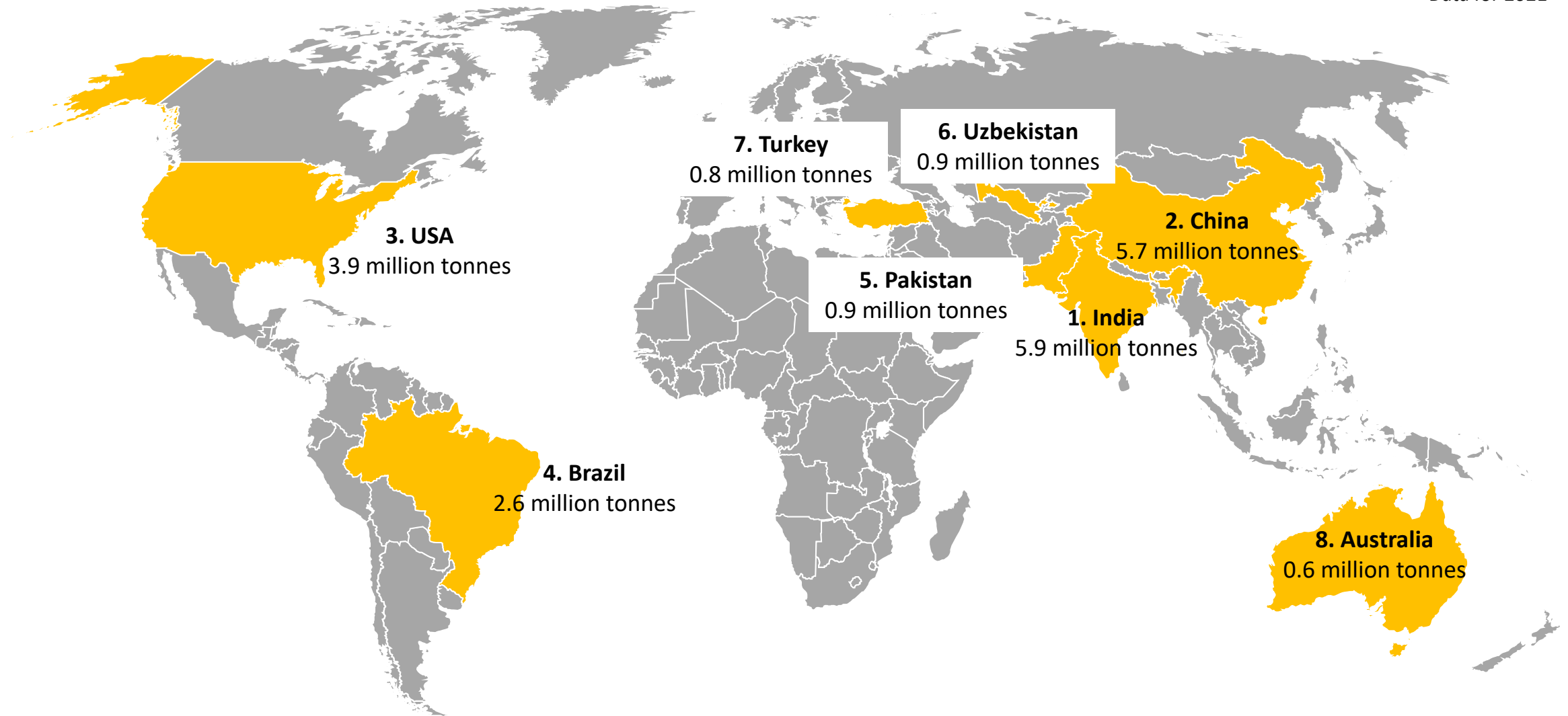
Values in million tonnes



Source: ICAC

Top 8 cotton producing countries account for approx. 85% of the global cotton production

Data for 2021



Source: ICAC

Cotton T-shirts and Jerseys form the largest share of globally exported cotton commodities

HS Code	Description	Trade Value (US\$ Bn)	Trade Share
610910	Knitted t-shirts of cotton	33.7	11%
611020	Knitted jerseys of cotton	30.2	10%
620342	Woven men's trousers, bib and brace overall, breeches and shorts of cotton	24.7	8%
620462	Woven women's trousers, bib and brace overall, breeches and shorts of cotton	22.0	7%
520100	Raw cotton fibre	16.9	6%
610462	Knitted men's trousers, bibs and brace overall, breeches and shirts of cotton	10.9	4%
620520	Woven men shirts of cotton	7.3	2%
611120	Knitted babies garments of cotton	7.1	2%
630260	Toilet linen and kitchen linen of terry fabrics of cotton	6.7	2%
610342	Knitted men's trousers, shorts of cotton	6.4	2%



Global Trends and their impact on CTA Industry



Manufacturers are actively looking to diversify out of China

China's apparel exports to US have reduced by 15% in the last decade

China's apparel exports to US have **continuously dropped** in the last decade.

This can be attributed to **US-China trade tensions**, diminishing cost advantage due to **increasing labour costs** and **growing geo-political distrust** between China and the West.

Row Labels	2010	2015	2019	2020	2021
China	40%	36%	30%	27%	25%
Viet Nam	8%	12%	16%	19%	17%
Bangladesh	5%	6%	7%	7%	8%
India	4%	4%	5%	4%	5%
Indonesia	6%	6%	5%	5%	5%
Cambodia	3%	3%	3%	4%	4%
Mexico	5%	4%	4%	3%	3%
Honduras	3%	3%	3%	3%	3%
Pakistan	2%	2%	2%	2%	3%
Nicaragua	1%	2%	2%	2%	2%
Row	21%	22%	23%	22%	23%
Total Imports	75.6	89.1	87.5	71.2	87.3

Xinjiang cotton issue caused several buyers to evaluate other options

Cotton from the Xinjiang region accounts for about **87% of the entire Chinese cotton and 20% of the entire world production.**

In late 2020, H&M and Nike boycotted Xinjiang cotton **over human rights concerns. Over the next few months, many other brands from US, UK, Canada, and EU followed suite.**

Tracing the origins of cotton is a complex task. Thus, most manufacturers in China are increasingly **preferring imported cotton fibre and yarn** for their products to circumvent the brands' reservations against Chinese cotton.

Source: UN Comtrade and Wazir Analysis

Stakeholders across the CTA value chain are pushing towards sustainability

Brands



Several large fashion brands and retailers have declared ambitious targets towards sustainable materials and practices

Consumers



Consumers' awareness of sustainable materials and trends is increasing.
Adoption of upcycled, recycled, repurposed, or reworked products is on rise

Policy makers



Sustainability mandates issued by authorities across the globe are pushing value chain stakeholders to adopt sustainable practices

*Sustainability based platforms such as **Common Objective** and **Fashion For Good** are also becoming increasingly relevant.*

Global Policymakers are Working Towards Legislations to Mandate Sustainability Principles in Fashion Value Chain

EU



- Launched strategy for sustainable and circular textiles in March 2022
- Policy targeting single-use plastics to collect 77% of beverage bottles by 2025 and 90% by 2029
- Imposed a plastics tax of 800 Euro/MT for increasing recycled content in the packaging industry
- ReHubs initiative to set up an integrated system based on 5 recycling hubs

California



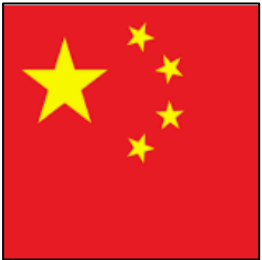
- Recently passed Senate Bill, requiring 25% reduction of single-use plastic packaging, all packaging to be recyclable or compostable, and a 65% plastic recycling rate – all of which are due by 2032
- Adopts a first-in-nation approach for protecting ocean and human health from microplastics pollution.
- Incentives and investments to businesses offering sustainable fashion services and thrift stores

Netherlands



- Green Deal Circular Procurement involves a €100 million to pilot circular procurement.
- A tax increase on natural gas and plans to price carbon emissions in combination with a reduction of income tax and employers' costs. (Sweden, Germany and Austria also)
- Low VAT for circular products and services influences the choice of consumers and businesses for circularity.

China



- Enacted environmental policies to crack down on textile dye pollution
- Beijing to promote low-carbon textile production, encourage using sustainable fibers, strengthen producers' social responsibility, improve the recycling system and raise investment in R&D for textile recycling technology

Increasing use of disruptive technologies

Hugo Boss's High Tech Factory in Turkey

Largest and most advanced HUGO BOSS production facility

- Area- 65,000 m²
- Products - Suits, jackets, shirts and coats
- Workers - 4,000 workers

The company demonstrates what Industry 4.0 looks like in practice here, with **networked machines, extensive data analysis and flexible processes.**

Digital transformation: Employees, machines & processes are networked to map & create a 'virtual twin' of the factory using more than 1,600 tablets tracking production data in real time



Robotics and automation: Absorption of new technologies & a 'TechnoLab' to develop semi- & fully automated machines & robots



Artificial intelligence: Factory data is collected to identify potential for improvement and risks. Digital forecasting for managing unit

Industry 4.0 Key Tools and Relevance in CTA Industry

Blockchain Technology

Using this technology and supply chain modelling, tracing every phase of garment making, every process used to get a product from point of origin to final retail destination is possible.

Artificial Intelligence

AI can access and collect historical and real-time operational data, providing insights that can improve operational efficiency

AR/VR (Augmented Reality/ Virtual Reality)

AR/ VR is mainly used to enhance the customer shopping experience with curated and individualized touches for each customer in the retail space.

Internet of Things (IoT)

IoT finds application in e-textiles, automated monitoring, predictive maintenance, increasing efficiency, product development

Focus on shorter supply chains

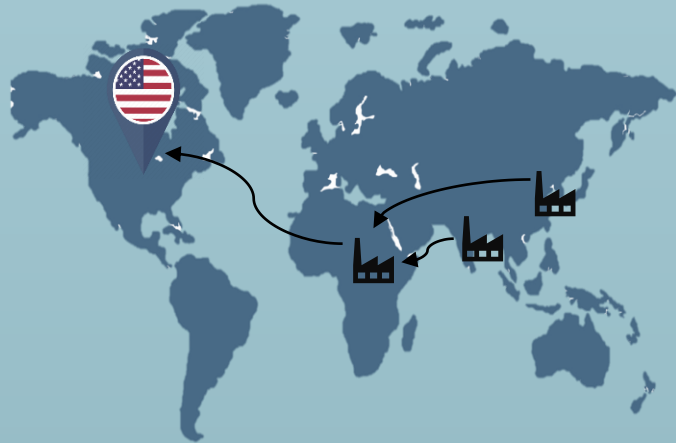
In order to shorten their supply chains, buyers are evaluating options like nearshoring and reshoring

Global Supply Chain Challenges

- Demand Volatility
- Logistics Jams
- Rising Costs
- Shipping Disruptions

Supply Chain Requirements

- High Agility & Control
- Shorter Lead Times
- Demand Driven-ness
- Lower Costs



Nearshoring

In order to de-risk their supply chain, lower operational costs and gain higher control, EU and US buyers are looking out to source from nearby suppliers Like Honduras, Turkey, Africa, etc.



Reshoring

Businesses are keen on shifting from a global supply chain model to one that is regionally based in order to increase supply chain efficiency.



Africa's CTA Industry Scenario



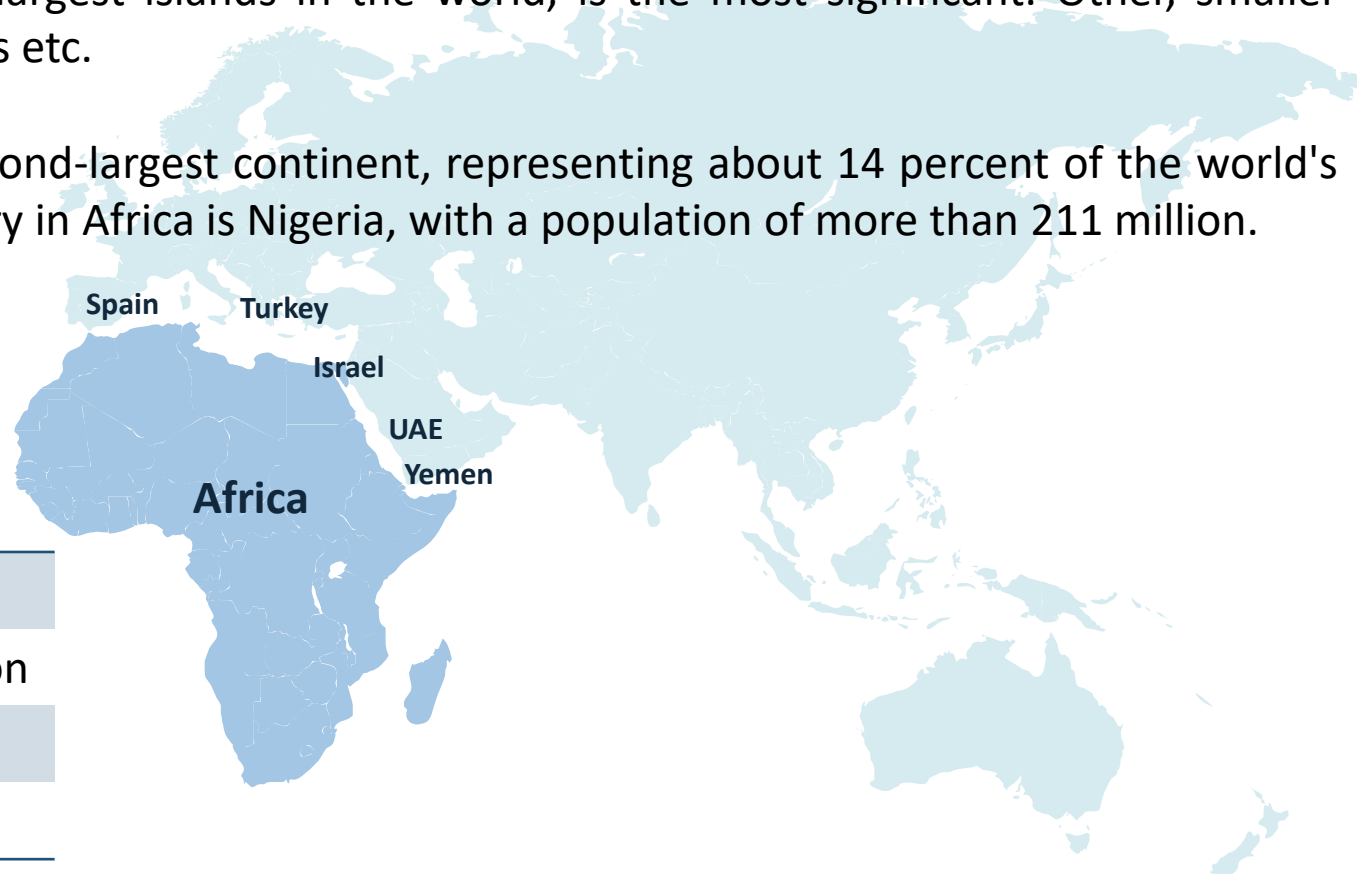
Demographics

Africa is the second largest continent in the world comprising of 50+ independent countries and a few disputed territories. Algeria is the largest country in the continent. Off the coasts of Africa a number of islands are associated with the continent - Madagascar, one of the largest islands in the world, is the most significant. Other, smaller islands include the Seychelles, Socotra, Comoros etc.

An estimated 1.37 billion people live in the second-largest continent, representing about 14 percent of the world's population (in 2021). The most populous country in Africa is Nigeria, with a population of more than 211 million.

Key Indicators for Africa (2022)

Population	1.4 billion
GDP (Current Prices)	US\$ 2.96 trillion
Real GDP Growth Rate	3.7%
GDP Per Capita (Current Prices)	US\$ 2,180

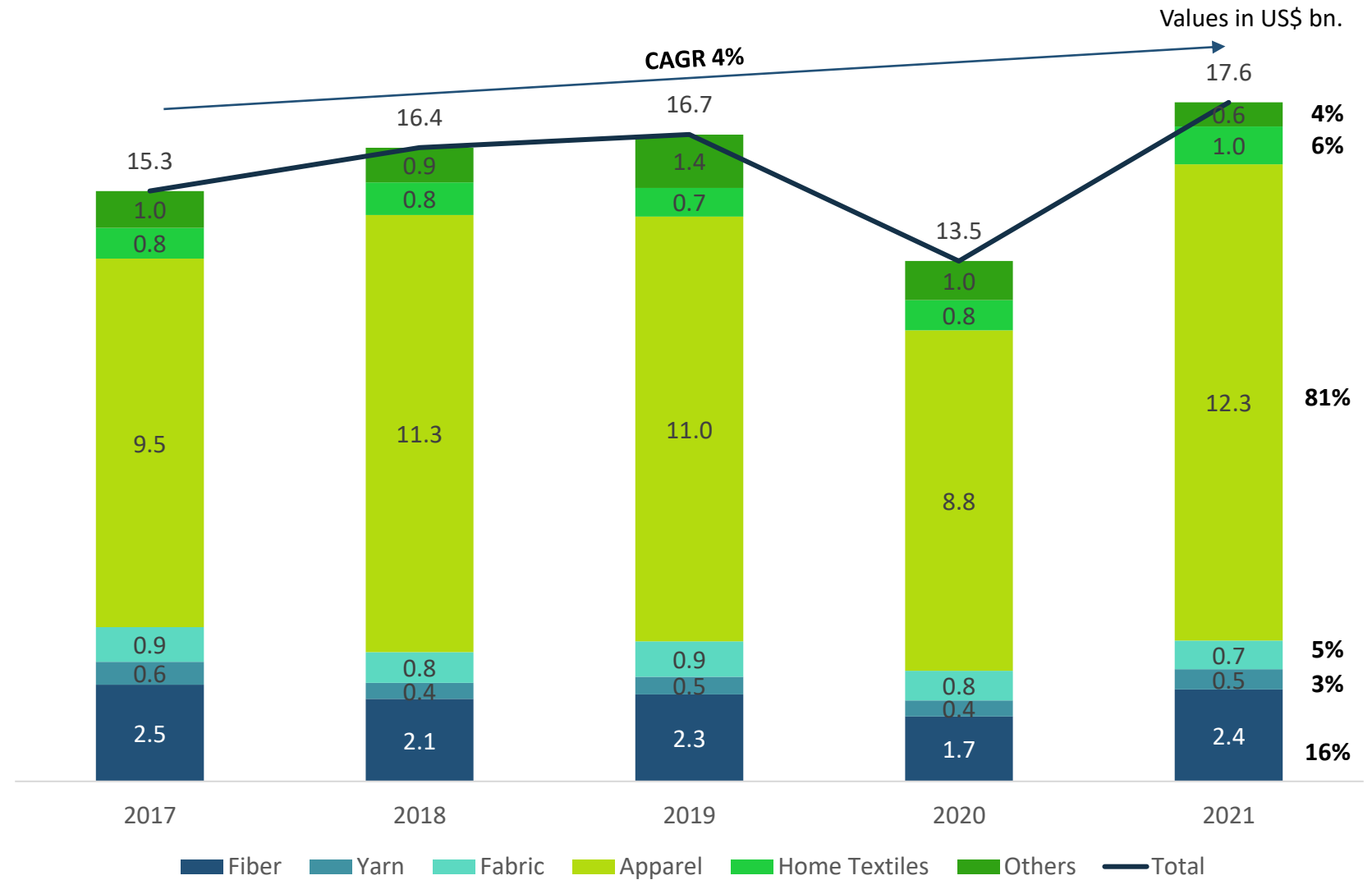


Data Source: IMF and UN

Africa makes for only 2% of the total global CTA exports

Africa's CTA exports have grown at a CAGR of 4% from 2017 to 2021, however, they make up a meagre share of 2% of the global exports.

Apparel is the largest exported category, followed by fibers, accounting for approx. 81% and 16% of Africa's exports in 2021, respectively.



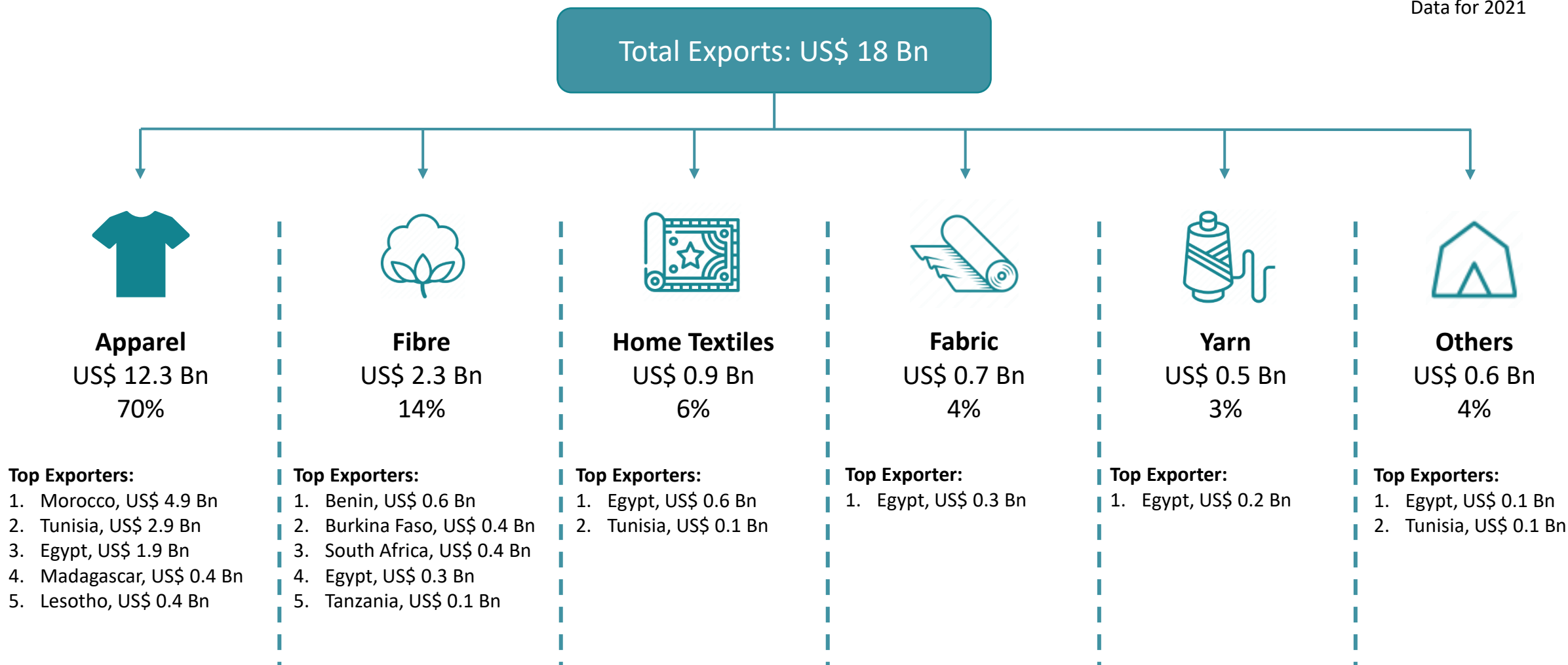
Source: UN Comtrade

North African countries account for the majority of African exports



Egypt and Tunisia are the largest exporters across categories

Data for 2021



Source: UN Comtrade

The largest exported commodities of Africa are cotton based

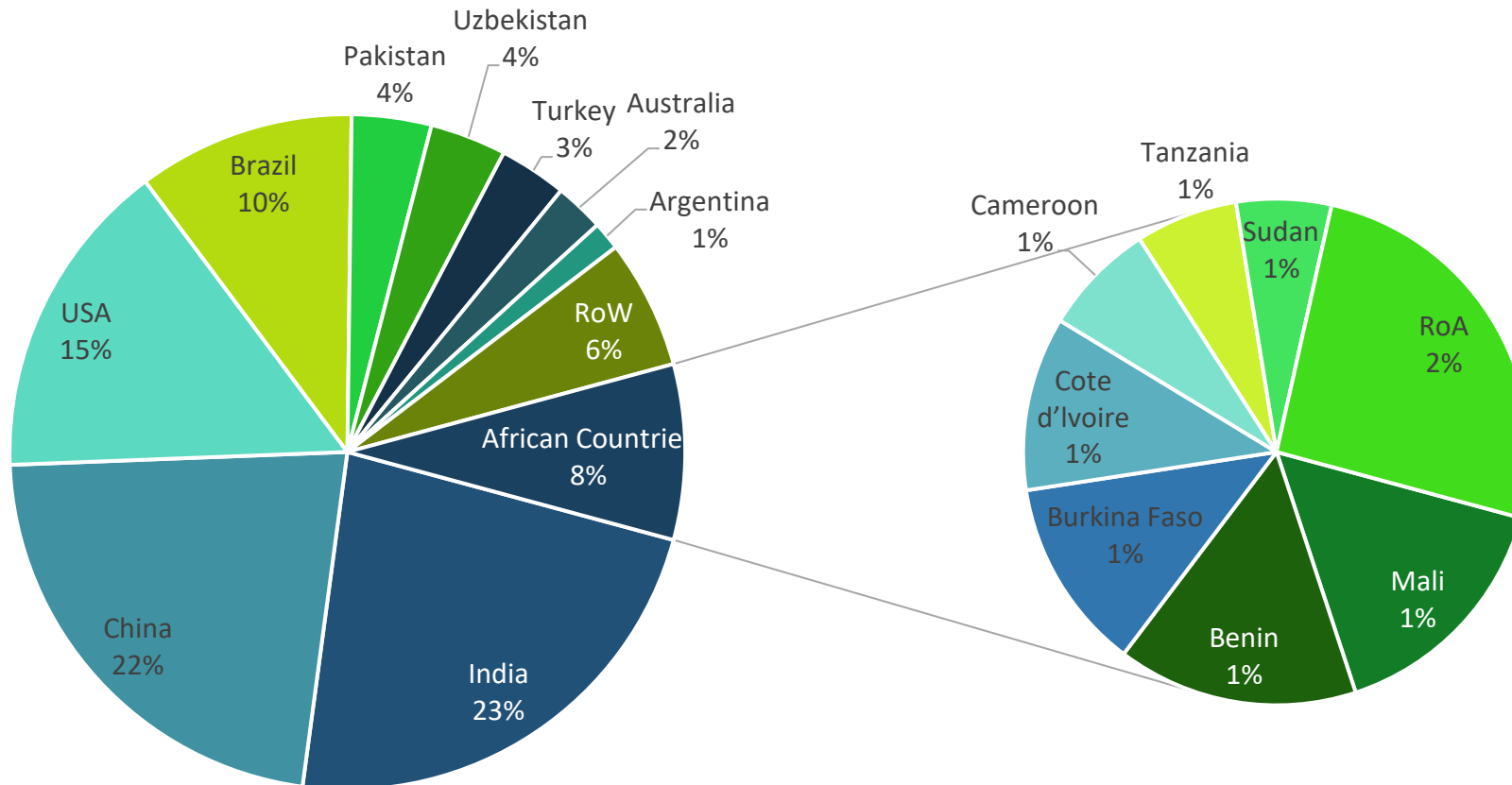
Data for 2021

HS Code	Description	Trade Value (US\$ Mn)	Trade Share
520100	Raw cotton fibre	1,638	9%
620342	Woven men's trousers bib and brace overalls breeches and shorts of cotton	1,588	9%
610910	Knitted t-shirts of cotton	860	5%
620462	Woven women's trousers bib and brace overalls breeches and shorts of cotton	766	4%
610990	Knitted t-shirts of fibers other than cotton	415	2%
620520	Woven men's shirts of cotton	337	2%
620343	Woven men's trousers bib and brace overalls breeches and shorts of cotton of synthetic fibres	324	2%
611020	Knitted jerseys of cotton	319	2%
620443	Woven women's dresses of synthetic fibres	306	2%
620640	Woven women's blouses, shirts of man-made fibres	291	2%

Source: UN Comtrade

African countries contribute to 8% of the total cotton production globally

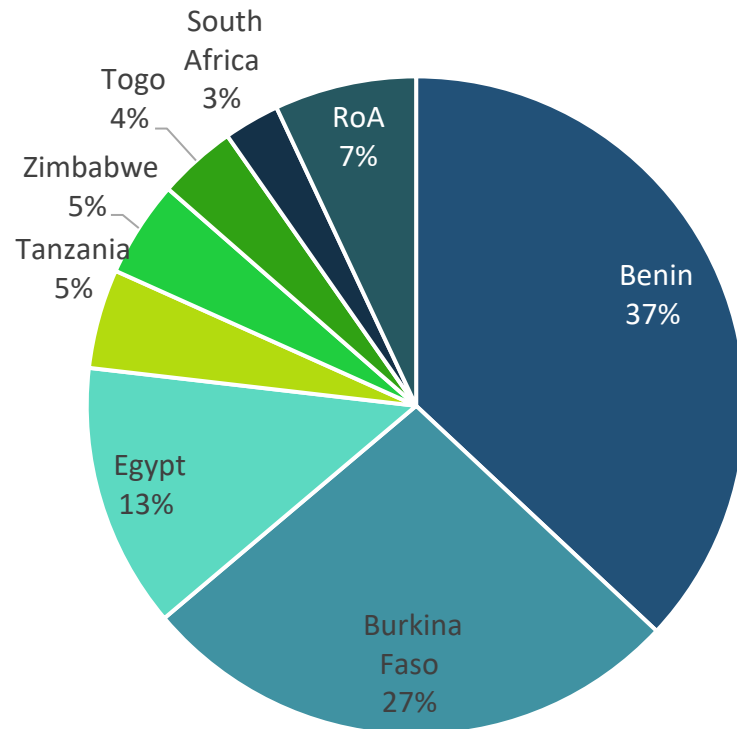
Data for 2021



African Countries	Production (000 tonnes)
Mali	340
Benin	332
Burkina Faso	265
Cote d'Ivoire	240
Cameroon	156
Tanzania	141
Sudan	131
Uganda	45
Rest of Africa	510
Total Production	2160

Data Source: ICAC

Top 7 cotton fibre exporters of Africa account for 93% of the exports

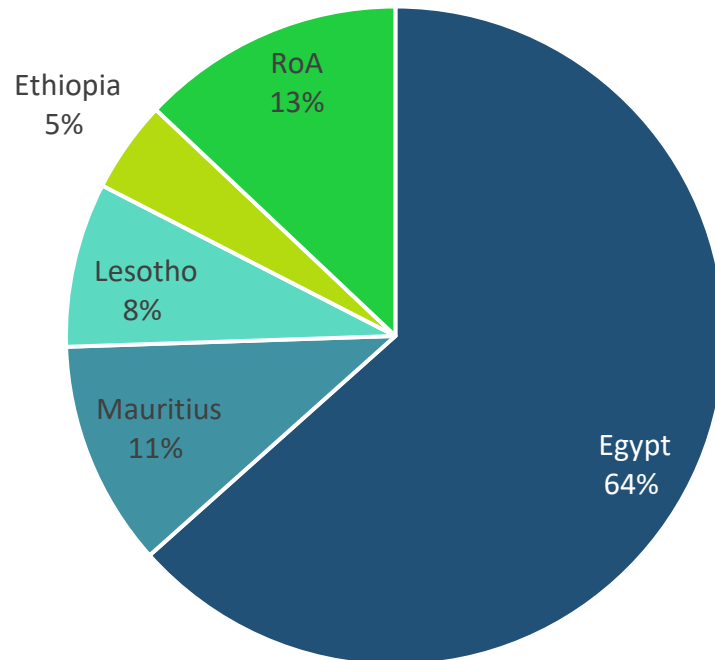


Value in US\$ mn.

Countries	2017	2021	CAGR (2017 to 2021)
Benin	347	626	16%
Burkina Faso	363	455	6%
Egypt	94	220	24%
Tanzania	38	83	22%
Zimbabwe	29	80	29%
Togo	73	65	-3%
South Africa	11	46	45%

Source: UN Comtrade

Top 4 cotton yarn exporters of Africa account for 87% of the exports

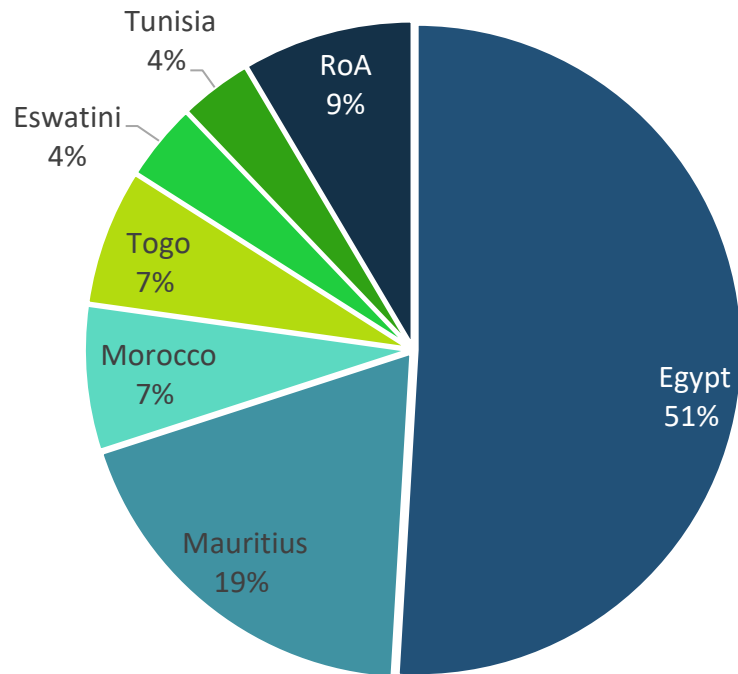


Value in US\$ mn.

Countries	2017	2021	CAGR (2017 to 2021)
Egypt	189	130	-9%
Mauritius	22	23	1%
Lesotho	19	17	-3%
Ethiopia	4	9	24%

Source: UN Comtrade

Top 5 cotton fabric exporters of Africa account for 91% of the exports

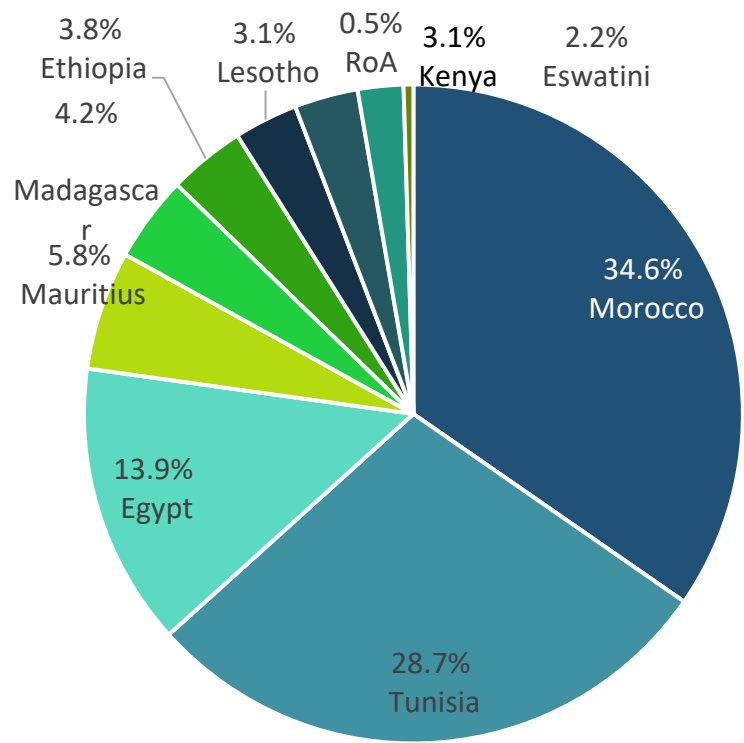


Value in US\$ mn.

Countries	2017	2021	CAGR (2017 to 2021)
Egypt	192	194	0%
Mauritius	48	73	11%
Morocco	45	27	-12%
Togo	1	26	122%
Eswatini	12	15	5%

Source: UN Comtrade

Top 9 cotton apparel exporters of Africa account for approx. 99% of the exports



Value in US\$ mn.

Countries	2017	2021	CAGR (2017 to 2021)
Morocco	1,097	2,000	16%
Tunisia	1,080	1,655	11%
Egypt	531	802	11%
Mauritius	486	337	-9%
Madagascar	205	243	4%
Ethiopia	73	218	31%
Lesotho	172	181	1%
Kenya	141	180	6%
Eswatini	109	129	4%

Source: UN Comtrade

Woven cotton apparel exports from Africa are higher than the knitted counterpart

Data for 2021

HS Code	Description	Trade Value (US\$ Bn)	Trade Share
620342	Woven men's trousers bib and brace overalls breeches and shorts of cotton	1,588	28%
610910	Knitted t-shirts of cotton	861	15%
620462	Woven women's trousers, bib and brace overalls, breeches and shorts of cotton	766	13%
620520	Woven men's shirts of cotton	337	6%
611020	Knitted women's jerseys of cotton	319	6%
620442	Woven women's dresses of cotton	186	3%
620630	Woven women's blouses, shirts and shirts-blouses of cotton	173	3%
611120	Knitted babies garments of cotton	157	3%
610462	Knitted women's trousers, bib and brace overalls, breeches and shorts of cotton	147	3%
610510	Knitted men's shirts of cotton	122	2%

Source: UN Comtrade

Africa has a strong base in cotton and related production

- Production in Egypt and the Sudan **more than doubled** between 2016/17 and 2018/19. Agriculture is irrigated in both countries and **temperatures** are extremely **uniform**. **Mill use** of cotton in Egypt is **nearly double** the amount of production and it continues to export **extra-fine cotton**.
- Cotton production and export in **Francophone Africa** rose rather significantly over the last couple of years representing about **2/3rd** of total cotton production in the region, with **area expansion**, rather than yields, underpinning production growth.
- Africa's lint exports are important for the livelihoods of millions of people, but this also underlines the commodity dependence of producing countries. Due to the **absence of technology** and the **technical know - how** of processing cotton linters, only **1.5%** of the cotton linters can be processed locally.
- Total ring spinning capacity has **recovered modestly** in Africa since 2017 from 3.5 million spindles to 3.9 million spindles in 2020. Investments in **forward integration** of cotton production has also picked up with installed shuttle less weaving machine capacity standing at 20,000 in 2020 from 19,000 in 2017.
- Sub Saharan Africa is a key supplier of **sustainable cotton** varieties. Top 5 countries including Cote d'Ivoire, Burkina Faso, Cameroon, Mali and Chad collectively produce about **0.6 million tons** of sustainable cotton which is **8%** of the total global production.

Source: FAO, UNCTAD, OTEXA, Eurostat, ITMF



Uganda's CTA Industry Scenario

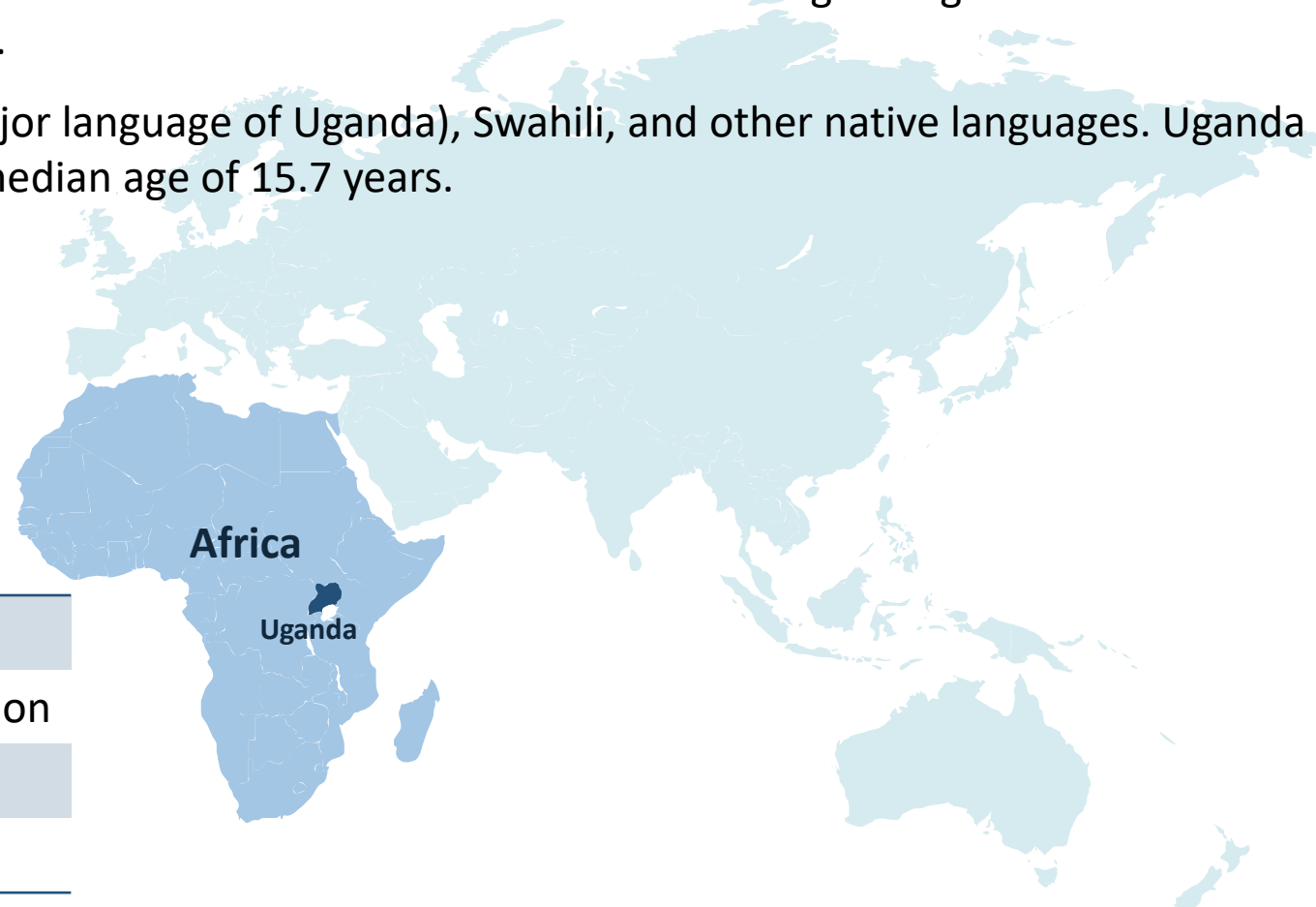


Demographics

- Uganda officially the Republic of Uganda is a landlocked country in East Africa. Uganda is one of the most densely populated countries in Africa with a population of 44 million. It is also one of the fastest growing countries in Africa. The largest city and capital is Kampala.
- Spoken languages are English, Luganda (a major language of Uganda), Swahili, and other native languages. Uganda has the world's youngest population with a median age of 15.7 years.

Key Indicators for Uganda (2022)

Population	44 mn.
GDP (Current Prices)	US\$ 48.35 billion
Real GDP Growth Rate	4.4%
GDP Per Capita (Current Prices)	US\$ 858



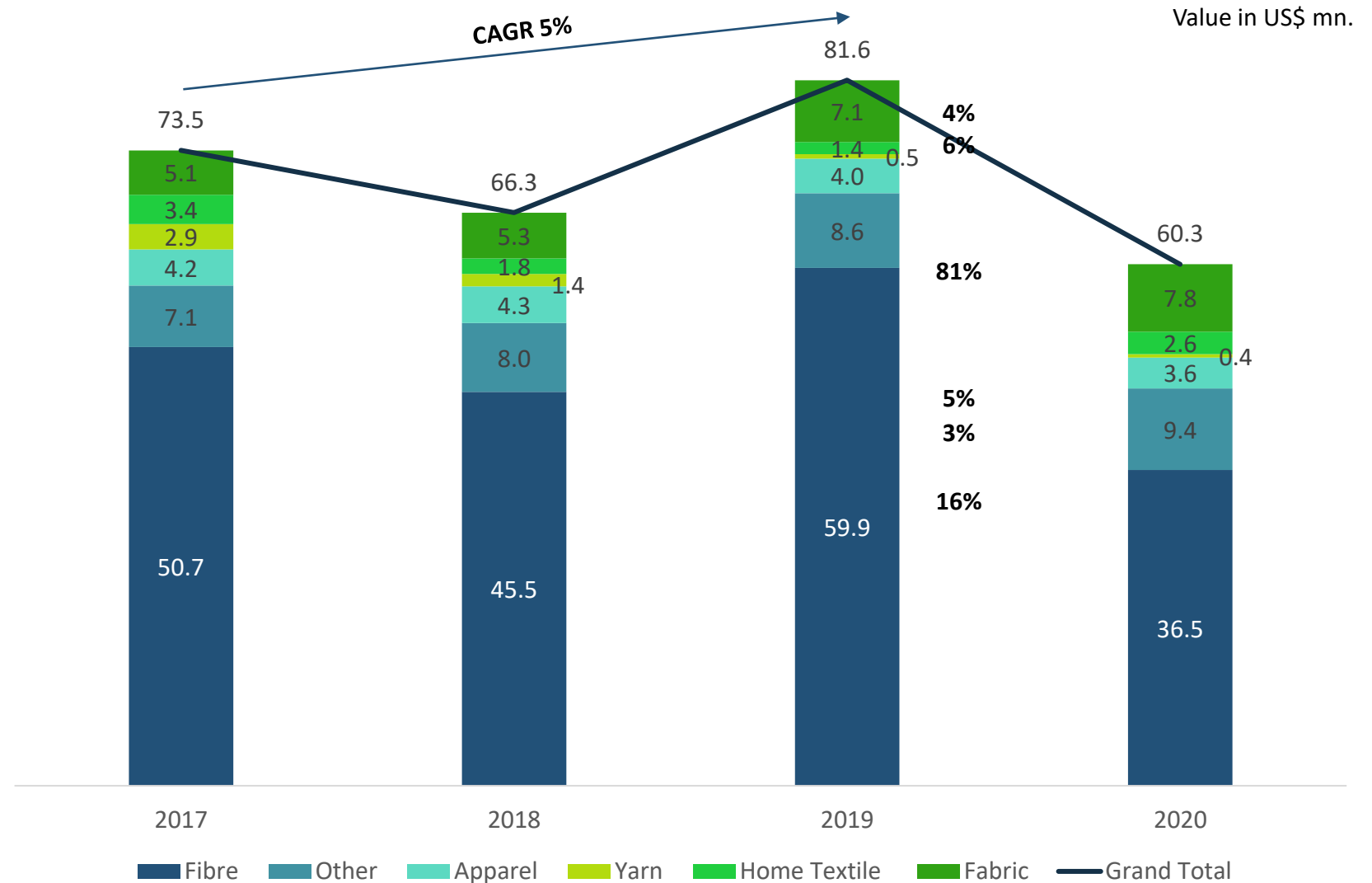
Data Source: IMF and UN

Fibre accounts for the highest share of Ugandan exports

Uganda's exports grew at a CAGR of 5% from US\$ 73.5 million in 2017 to US\$ 81.6 million in 2019.

Fibre is the largest exported commodity over the years, maintaining an average share of 68% from 2017 to 2020.

The share of fibre exports went to as high as 71% in 2019.



Source: UN Comtrade

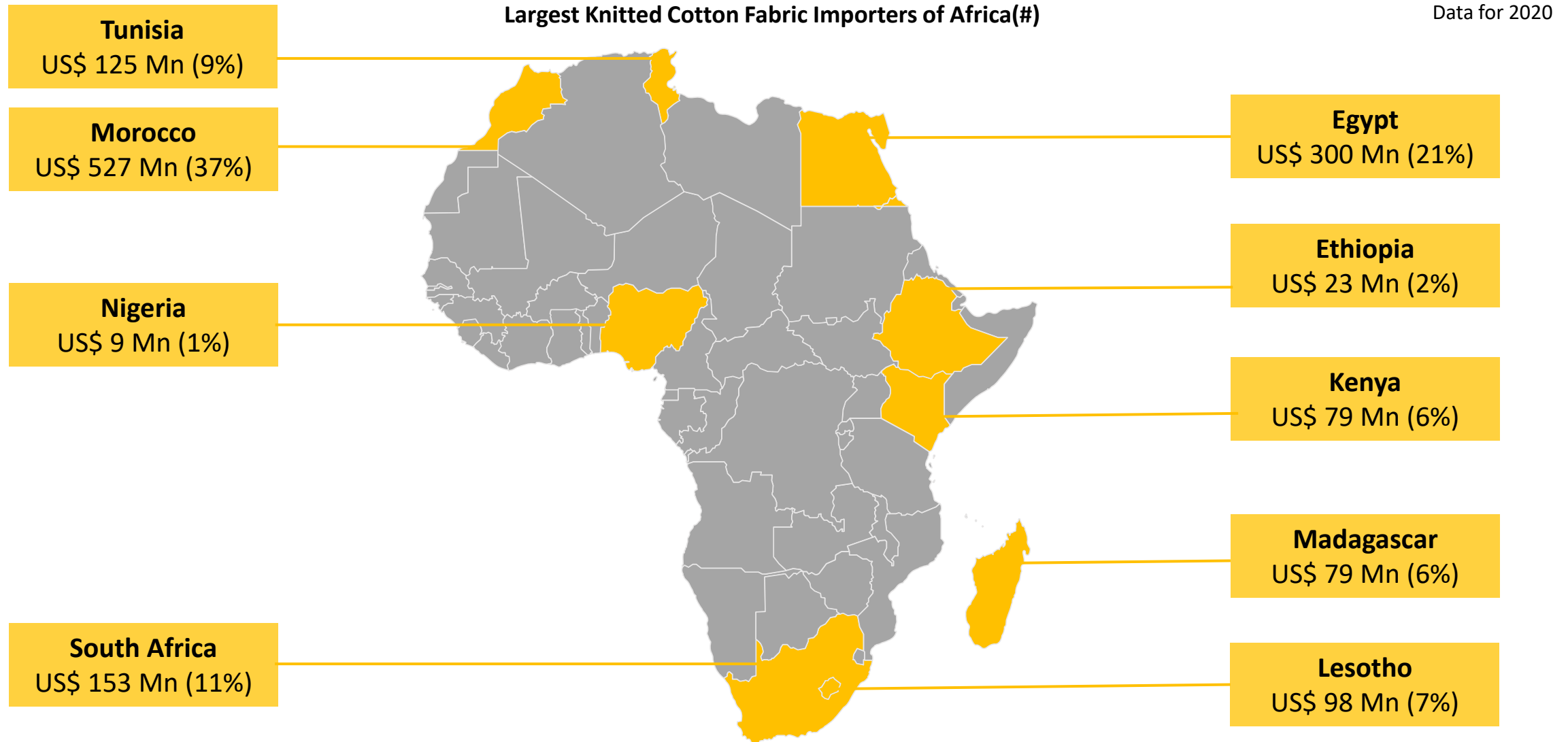
Cotton fibre accounts for approx. 58% of the total exports from Uganda

Data for 2020

HS Code	Description	Trade Value (US\$ Mn)	Trade Share
5201 & 5203	Cotton fibre	34.5	58%
630533	Sacks/Bags of polyethylene or polypropylene strip	3.5	6%
630640	Pneumatic Mattresses of miscellaneous fibers	3.0	5%
540751	Bleached/Unbleached woven fabric containing texturized polyester filaments	1.9	3%
521049	Yarn Dyed Cotton Fabrics <200 GSM	1.8	3%
540744	Woven printed fabrics of Nylon Filament Yarn	1.6	3%
621010	Non-Woven garments of any fibre	1.2	2%

Source: UN Comtrade

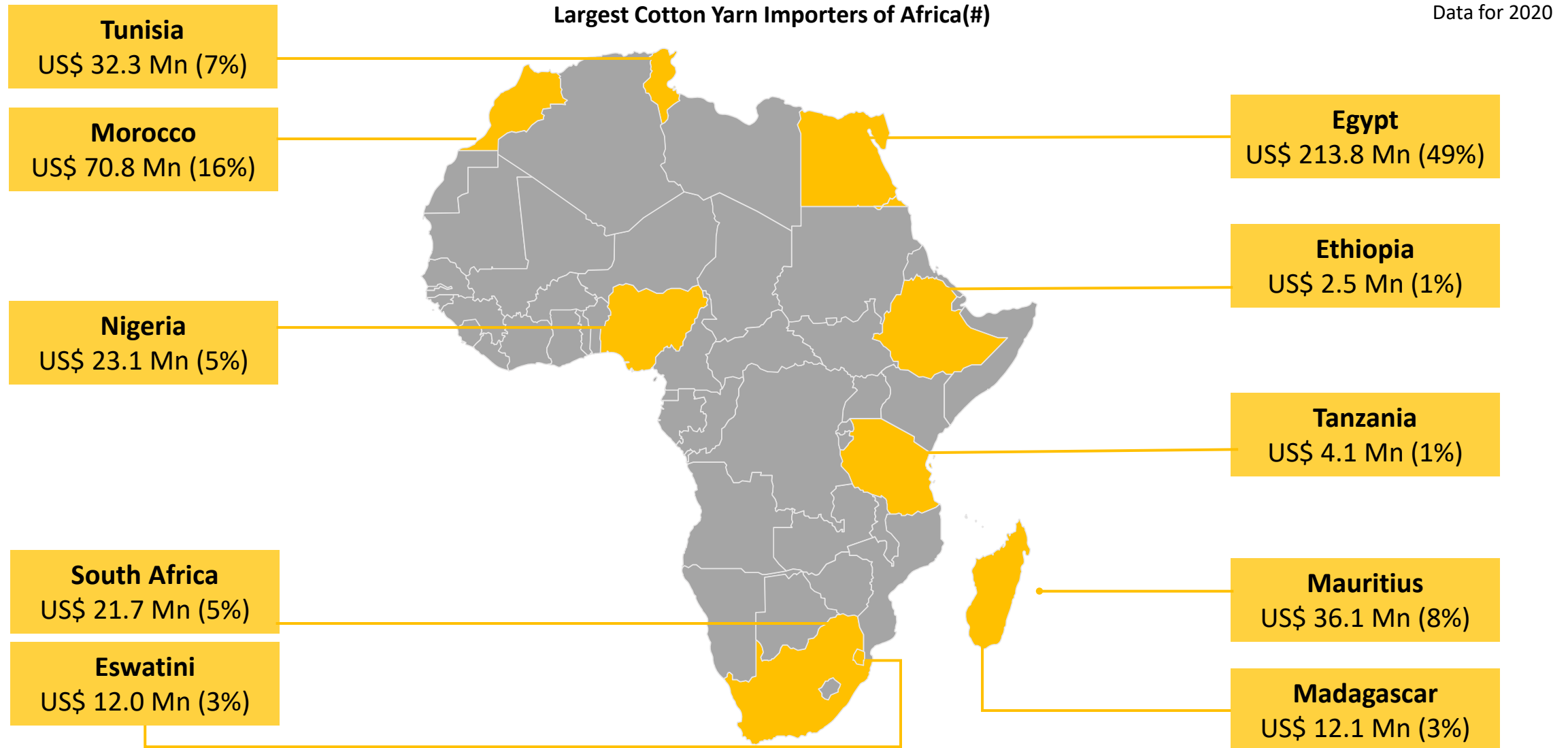
Regional knitted cotton fabric importers can serve as a market for a fabric mill set up in Uganda



Source: UN Comtrade

(#) - Percentage share of total African knitted cotton fabric imports

Regional cotton yarn importers can also serve as a market for a spinning mill set up in Uganda



Source: UN Comtrade

(#) - Percentage share of total African cotton yarn imports

Uganda needs industry intervention as well as government support in order to develop manufacturing competitiveness

Cost element	Unit	Uganda	Morocco	Tunisia	Egypt	Kenya	Ethiopia
Power Cost	US\$ / kwh	0.14	0.10	0.10	0.06	0.17	0.02
Labour Cost	US\$ / month	150	285 - 330	120 - 140	95 - 105	135-175	46
Lending Rate	%	18 - 20	4.2 - 6	7.5 - 8	14 - 15	14	12 - 15

Source: Gatsby - Country Benchmarking Report (March 2022), Global Petrol Prices, CEIC, Industry Feedback

Setting up an integrated fabric mill in Uganda: Opportunities & Challenges

Opportunities

- Easy availability of raw cotton
- Availability of ready to work inexpensive labour
- Duty free access to the leading global markets of US and EU
- Intra-Africa trade provisions under AfCFTA, COMESA, EAC, etc.
- Government support through launch of favourable policies like National Cotton Textile and Apparel Strategy (NCTAS)

Challenges

- Uganda being a landlocked country faces logistical limitations to ship across markets
- Unavailability of skilled and experienced workforce leading to lower productivity
- Consumer inclination towards second hand garments, hampering sales growth in domestic market
- High interest rates
- Unavailability of PSF
- Limited availability of machinery as well as spare parts and service issues in case of breakdowns
- Fluctuations in cotton prices



Raw Material Sources

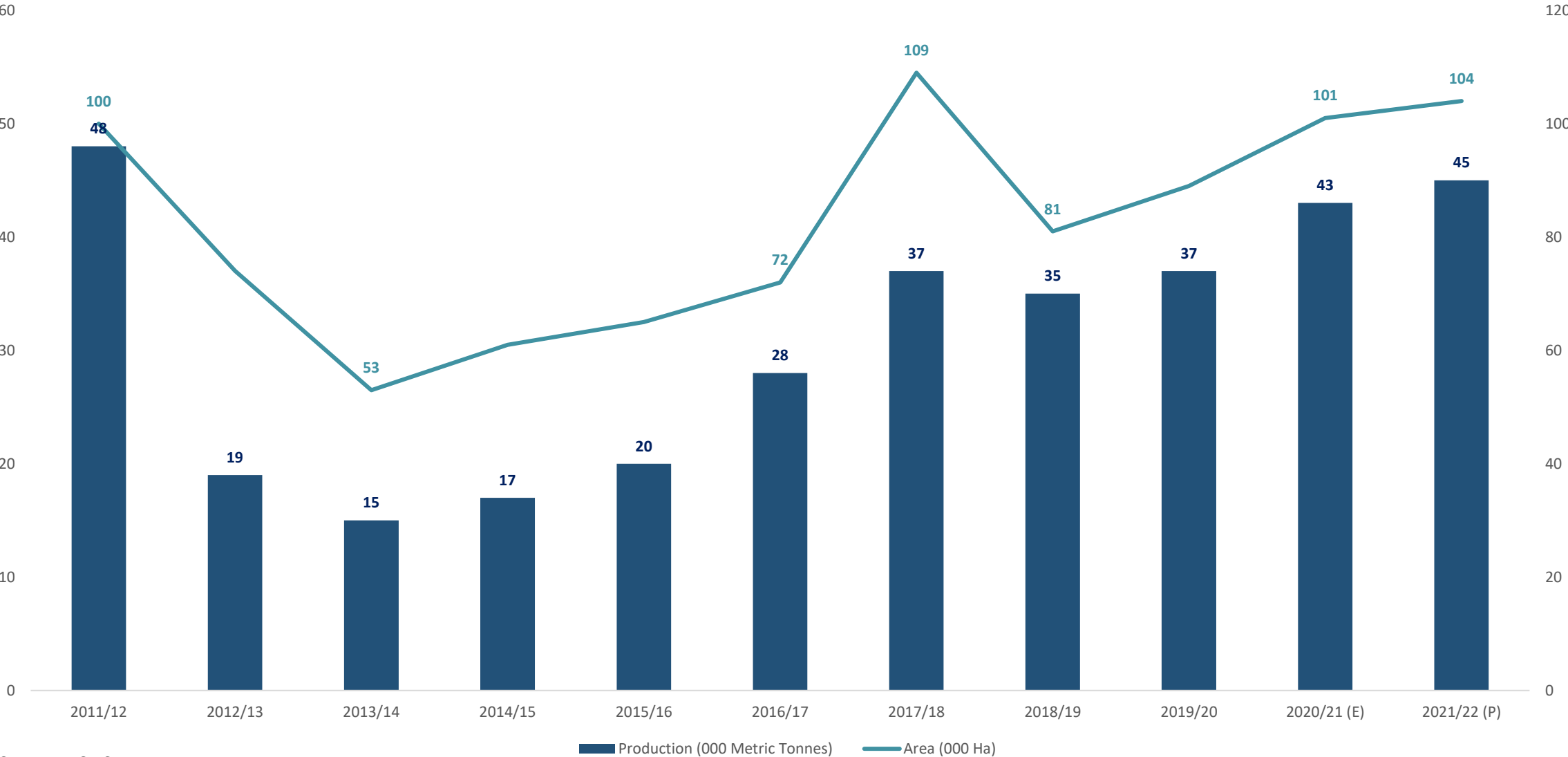




Cotton

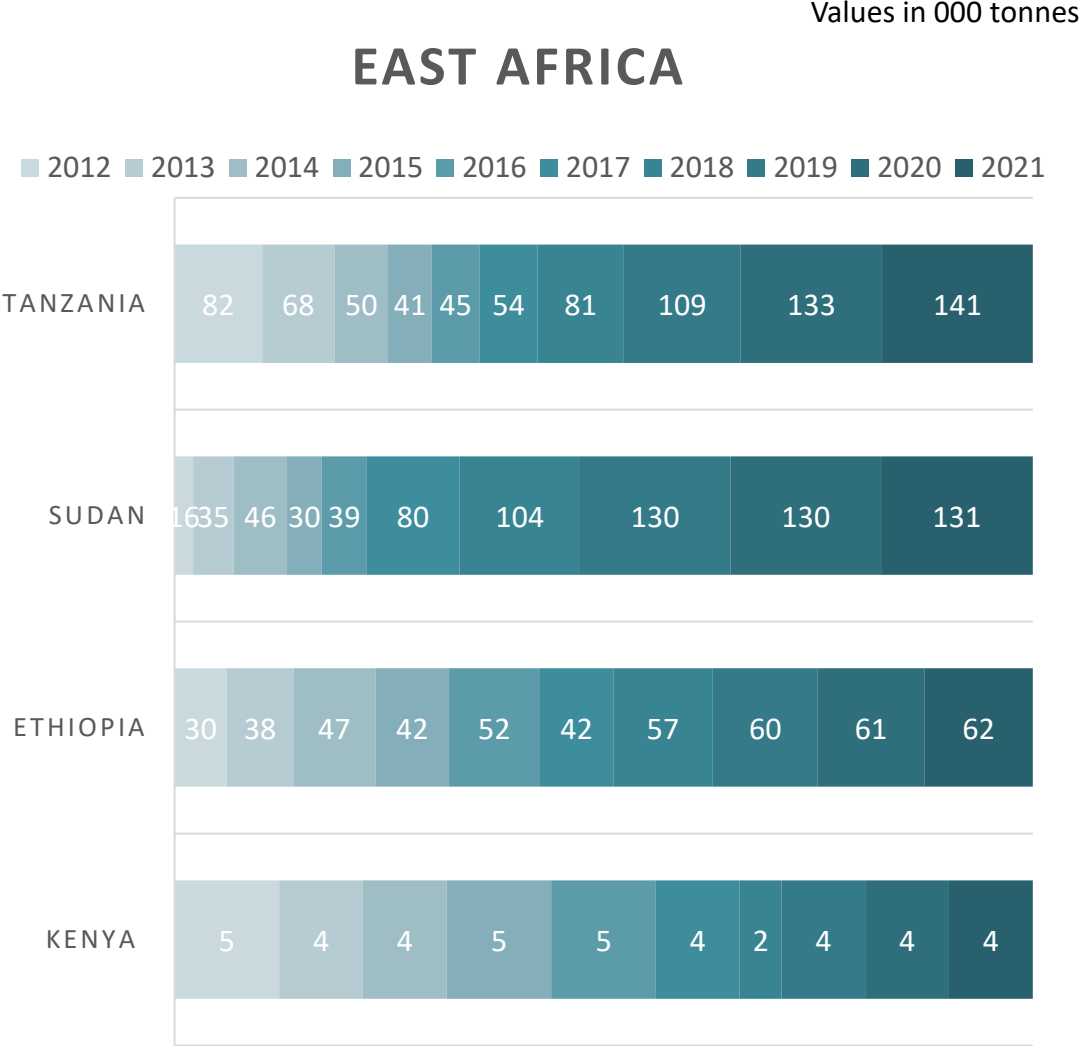
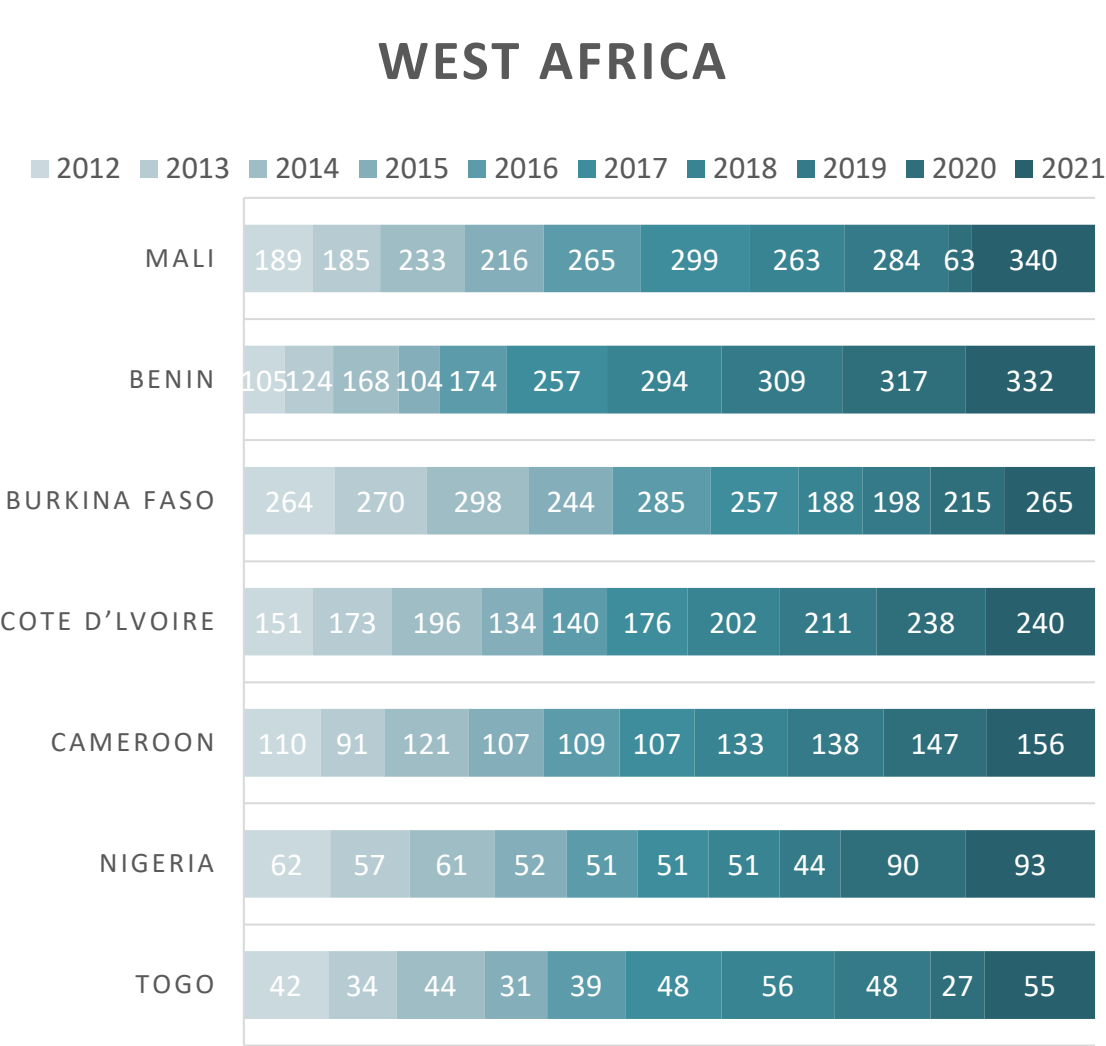


Uganda produces ample cotton fibre in order to cater to the raw material requirement of a spinning mill



Source: ICAC

Cotton fiber can also be imported from leading cotton producers of West & East Africa



Source: ICAC

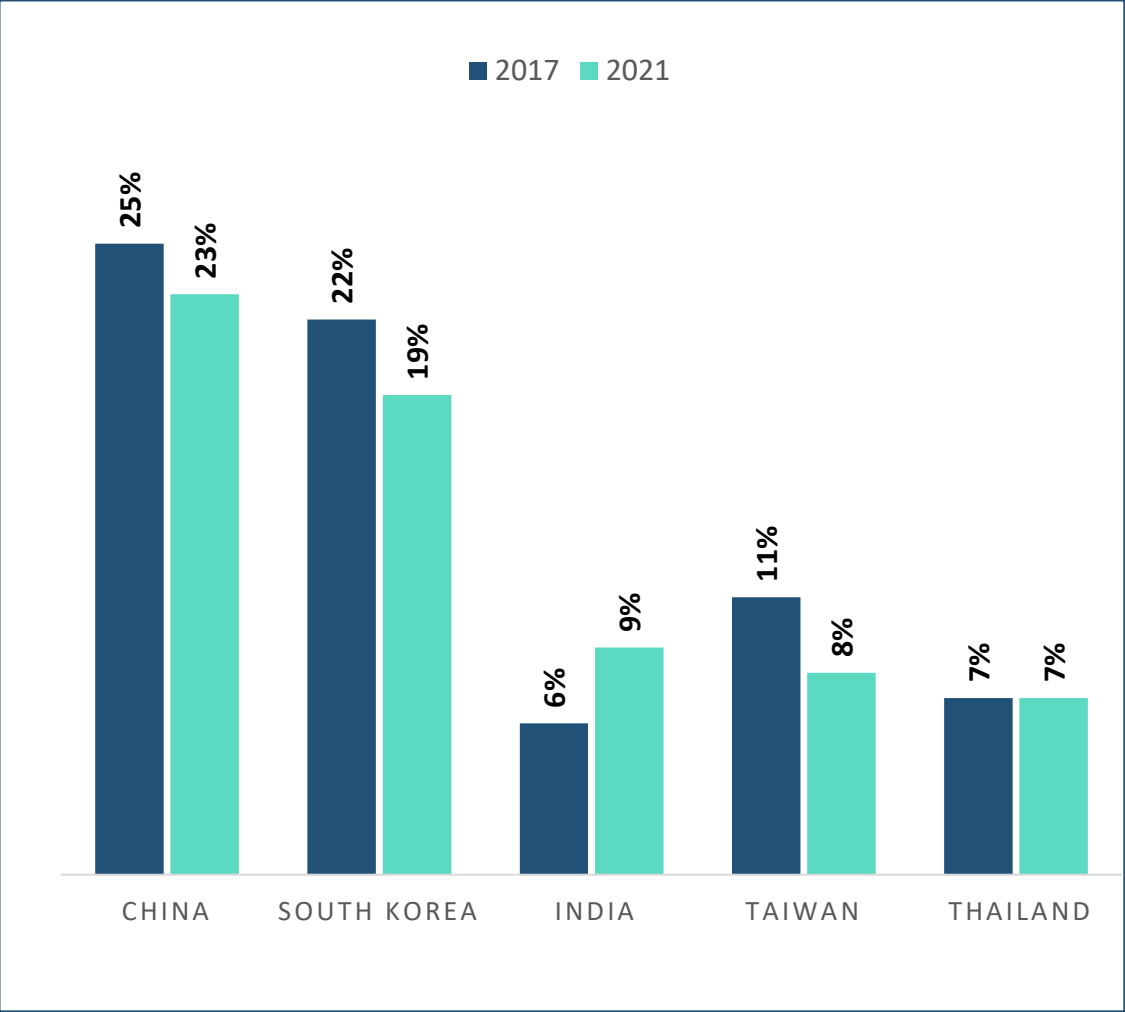


Polyester



Polyester can be sourced from leading as well as emerging global exporters

Major Exporters & their Share Change



Source: UN Comtrade

Emerging Exporters & their Growth Rate

Emerging Exporters*	Exports 2017 (US\$ mn.)	Exports 2021 (US\$ mn.)	CAGR
Turkey	49	193	41%
Viet Nam	75	200	28%
India	244	411	14%
Indonesia	159	252	12%

- Uganda can source PSF from global leading exporters like China, South Korea, India, Taiwan and Thailand.
- PSF can also be sourced from emerging exporters like Turkey, Vietnam and Indonesia which are growing at double-digit CAGRs since 2017.

*Fastest growing exporter with minimum exports of US\$ 100 mn.

Leading Manufacturers of PSF

Key Manufacturers	2021 Capacities (ktpa)	Manufacturing Locations
Yizheng Chemical Fibre (Sinopec)	1,100	China
Jiangsu Sanfangxiang	800	China
Reliance Industries Limited	892	India
Fujian Jinlun	600	China
SASA	510	Turkey
Huvis	396	S. Korea
Indorama Polyester	370	Indonesia
Indorama Synthetics	300	India
Toray Advanced Material	261	S. Korea
Nan Ya	220	USA
Asia Pacific Fibres	195	Indonesia
Bombay Dyeing	165	India
DAK Americas	150	USA
PVTEX	144	Vietnam
Akinal	140	Turkey

Source: PCI Fibres

Key Takeaways

- The global emerging trends of manufacturing diversification out of china, focus on shorter supply chains as well as emphasis on sustainability across the value chain create a huge opportunities for the African countries to tap the international markets.
- Cotton garments are the largest exported commodities globally and from Africa, while more than 50% of Uganda's exports account for cotton fibres. This highlights the opportunity to set up manufacturing units in Uganda for value addition in the raw fibre.
- Uganda's cotton production has been increasing over the years, which can be well utilized in the proposed spinning mill.
- Raw material can also be sourced from other cotton producing African countries like Mali, Benin, Burkina Faso, Ivory Coast, Tanzania, Sudan, etc.
- Cotton Yarn's produced by the Ugandan spinning mill can substitute imports in the African continent. Countries like Egypt, Morocco, Tunisia, Nigeria, Tanzania, Ethiopia that import cotton yarns can serve as markets for the same. However, this will require Uganda to become competitive against the Asian cotton yarn suppliers. This can be done by the amalgamation of right technology, skilling and training of workforce and government support.
- Setting up an integrated fabric mill in Uganda comes with its own sets of opportunities and challenges. The potential challenges can be mitigated with the combined efforts of industry stakeholders, investors and the Ugandan government.
- Forward integration to garment manufacturing is also an attractive opportunity that can be considered.



End

